

MUST lecture 3

One-sample Hotelling T-squared test

Inference for mean vector for single MVN sample

Disclaimer:

**Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with
MATLAB examples**

One-sample MVN model

$$X_1, X_2, \dots, X_n \text{ iid } X_j \sim N_p(\mu, \Sigma)$$

Null hypothesis

$$H_0: \mu = \mu_0 \quad \text{against} \quad H_1: \mu \neq \mu_0$$

Point estimators

$$\hat{\mu} = \bar{X} = \frac{1}{n} \sum_{j=1}^n X_j$$
$$\hat{\Sigma} = S = \frac{1}{n-1} \sum_{j=1}^n (X_j - \bar{X})(X_j - \bar{X})^T$$

Test statistic and distribution

$$T^2 \stackrel{\text{def}}{=} (\bar{X} - \mu)^T \left(\frac{S}{n} \right)^{-1} (\bar{X} - \mu) \sim \frac{p(n-1)}{n-p} F(p, n-p)$$

Test

Put in estimators and H_0 : $T^2 = t^2$

1) choose level of significance α

$$\text{reject } H_0 \Leftrightarrow t^2 > \frac{p(n-1)}{n-p} F(p, n-p)_\alpha$$

2) calculate p-value = $P\left(F(p, n-p) > \frac{n-p}{p(n-1)} t^2\right)$

reject H_0 for $\forall \alpha > p\text{-value}$

Confidence region for unknown mean vector

$(1 - \alpha)100\%$ confidence region for μ

$$\forall \mu \in \mathbb{R}^p: (\bar{X} - \mu)^T \left(\frac{S}{n} \right)^{-1} (\bar{X} - \mu) \leq \frac{p(n-1)}{n-p} F(p, n-p)_\alpha$$

p -dimensional (hyper)ellipsoid (can not be visualized for $p > 3$)

Marginal confidence intervals for mean vector components (too narrow)

$(1 - \alpha)100\%$ marginal CI for $\mu_i, i = 1, \dots, p$

$$\left[\bar{x}_i \pm t(n-1)_{\alpha/2} \sqrt{\frac{S_{ii}}{n}} \right], \quad i = 1, \dots, p$$

Bonferroni confidence intervals for mean vector components (compromise)

$$\left[\bar{x}_i \pm t(n-1)_{\alpha/2p} \sqrt{\frac{S_{ii}}{n}} \right], \quad i = 1, \dots, p$$

Simultaneous confidence intervals for mean vector components (too wide)

$$\left[\bar{x}_i \pm \sqrt{\frac{p(n-1)}{n-p} F(p, n-p)_\alpha} \sqrt{\frac{S_{ii}}{n}} \right], \quad i = 1, \dots, p$$

Large sample approximate inference for mean vector

For $n \gg p$ (large sample):

Approximately $(\bar{X} - \mu)^T \left(\frac{S}{n}\right)^{-1} (\bar{X} - \mu) \sim \chi^2(p)$

More important, because of the Central Limit Theorem this is *true even if individual observations are not from a multivariate normal distribution*.

CR and CI's based on large sample approximation

Confidence region for μ : $(\bar{X} - \mu)^T \left(\frac{S}{n}\right)^{-1} (\bar{X} - \mu) \leq \chi^2(p)_\alpha$

Marginal CI's $\left[\bar{x}_i \pm z_{\alpha/2} \sqrt{\frac{S_{ii}}{n}} \right], \quad i = 1, \dots, p$

Bonferroni CI's $\left[\bar{x}_i \pm z_{\alpha/2p} \sqrt{\frac{S_{ii}}{n}} \right], \quad i = 1, \dots, p$

Simultaneous CI's $\left[\bar{x}_i \pm \sqrt{\chi^2(p)_\alpha} \sqrt{\frac{S_{ii}}{n}} \right], \quad i = 1, \dots, p$

Likelihood Ratio Test for mean vector

Given observations $X_1, X_2, \dots, X_n$ iid $X_j \sim N_p(\mu, \Sigma)$

Unconstrained likelihood function $L(\mu, \Sigma) \stackrel{\text{def}}{=} P(\text{given observations} \mid \mu, \Sigma) = \prod_{j=1}^n P(X_j \mid \mu, \Sigma)$

H_0 likelihood function $L(\mu_0, \Sigma) \stackrel{\text{def}}{=} P(\text{given observations} \mid \mu = \mu_0, \Sigma) = \prod_{j=1}^n P(X_j \mid \mu = \mu_0, \Sigma)$

Likelihood Ratio $\Lambda_{LR} \stackrel{\text{def}}{=} \frac{\max_{\Sigma} L(\mu_0, \Sigma)}{\max_{\mu, \Sigma} L(\mu, \Sigma)}, \quad 0 \leq \Lambda_{LR} \leq 1$

Put MVN pdf in likelihood functions: $\Lambda_{LR} = \left(\frac{\hat{\Sigma}}{\hat{\Sigma}_0} \right)^{n/2}$ where

$$\hat{\Sigma} = \frac{1}{n} \sum_{j=1}^n (X_j - \bar{X})(X_j - \bar{X})^T$$
$$\hat{\Sigma}_0 = \frac{1}{n} \sum_{j=1}^n (X_j - \mu_0)(X_j - \mu_0)^T$$

Large sample approximate LR test for $H_0: \mu = \mu_0$: $-2 \log \Lambda_{LR} \sim \chi^2(df - df_0)$ where
 $df = \dim(L(\mu, \Sigma)), \quad df_0 = \dim(L(\mu_0, \Sigma))$